

Joint 3-class Lung Segmentation — SegResNet, ex-LIDC (NLST + NSCLC only) — sliding-window (128^3)

1. Task

End-to-end 3-class semantic segmentation of a full chest CT into {background, lung, nodule}. Same task as the paper's `joint_segresnet_ex_lidc` — the only difference is inference recipe: **sliding-window over 128^3 patches at native resolution** rather than a single forward on a 256^3 resample.

- **Input** (1, H, W, D) CT at native voxel grid.
- **Training input** 128^3 patches sampled via `RandCropByLabelClassesd` with class-balanced ratios `[bg, lung, nodule] = [0.2, 0.3, 0.5]`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, output at input resolution.
- **Output** (3, H, W, D) softmax logits: 0 = bg, 1 = lung, 2 = nodule.

2. Model

Property	Value
Architecture family	SegResNet
Spatial dims	3
Input channels	1
Output channels	3 (softmax)
Init. feature width	16
Trainable parameters	20,663,555

3. Data

Source	Role
NLST	train + val + test
NSCLC-Radiomics	train + val + test
LIDC-IDRI	excluded (no GT lung labels available)

- **Split**: `unified_ex_lidc.json` — patient-grouped, dataset-stratified.
- **Preprocessing**: none beyond intensity clipping / normalization; the full CT enters the sliding-window inference at its native voxel grid.

4. Training

Setting	Value
Loss	Multiclass Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1$, class weights [1, 1, 100])
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	2 volumes $\times$ 4 patches = 8 patches / step
Patch size	128^3 (class-biased random crop, ratios [bg, lung, nodule] = [0.2, 0.3, 0.5])
Epochs	400
Random seed	42
Mixed precision	bf16
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 $\times$ NVIDIA H100 80 GB (fat01)

Best checkpoint recorded at **epoch 349 / 400**.

5. Evaluation protocol

Evaluated on the `unified_ex_lidc` **test** split (129 series). All voxel-level metrics micro-averaged over the whole split. Per-class metrics computed against argmax predictions.

6. Results

Ticket-required metrics (3-class average)

Metric	Value
mean IoU (3-cls)	0.7627
mean IoU (lung + nodule)	0.6458
Accuracy	0.996442
Macro Precision (lung + nodule)	0.6649
Macro Recall (lung + nodule)	0.9347

Per-class breakdown

Class	Dice (micro)	Recall	Precision	IoU
background	0.9982	0.9979	0.9985	0.9964
lung	0.9789	0.9747	0.9831	0.9587
nodule	0.4996	0.8946	0.3466	0.3330

Per-case Dice distribution — nodule class (129 test cases)

Statistic	Value
Mean	0.4447
Std	0.2498
Min	0.0000
p25	0.2207
Median	0.4964
p75	0.6303
Max	0.8865

Voxels evaluated: 9,789,871,397. Cases: 129.

7. Notes on interpretation

Lung Dice at ≈ 0.98 is essentially saturated across all architectures and inference recipes — the lung is a large, well-defined foreground class. Nodule Dice is the real signal.

Sliding-window inference at 128^3 trades global lung context for preserved local resolution. On this corpus the trade tends to be neutral-to-negative on nodule Dice compared to the 256^3 resize baseline — patches see only $\sim 15\%$ of a typical lung volume, which limits anatomy-conditional reasoning.

8. Reproducibility

- Config `/home/werner/nodule-training-demo/configs/joint_segresnet_ex_lidc_sw.yaml`
- Checkpoint `stage3_joint/checkpoints_segresnet_ex_lidc_sw/best_model.pth`
- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/sw_metrics/joint_segresnet_ex_lidc_sw.json`
- Command `python stage3_joint/eval_metrics.py --config /home/werner/nodule-training-demo/configs/joint_segresnet_ex_lidc_sw.yaml --checkpoint stage3_joint/checkpoints_segresnet_ex_lidc_sw/best_model.pth --split test`